

FREEZE-DRIED FOOD

Although the process of freeze-drying to preserve food was invented in the 1930s, its unique benefits were discovered by NASA scientists looking at methods for storing food on long-duration Apollo spaceflights. Freeze-drying removes water but preserves vitamins, minerals, and other nutrients that are restored when the meal is rehydrated.



Fruit retains its appetizing color and flavor

Freeze-dried strawberries

WOW!

According to some estimates, each dollar invested in the US space program results in up to \$10 of wider long-term benefits.

FAST FACTS

- Although the principle of generating electricity from sunlight was discovered in the 19th century, cheap and efficient solar cells originated in a NASA-sponsored project to develop solar-powered aircraft.
- Famous inventions that are often wrongly attributed to the space race between the US and Russia include Teflon™, Velcro®, and the integrated circuit (microchip).

SPACE

FIRE SAFETY

Fire is one of the deadliest hazards for spacecraft and their crews, and space agencies such as NASA have made some important breakthroughs in fire safety. Firefighters use air bottles, face masks, and other equipment made from a heat-resistant aluminum composite that was first developed for use in rocket casings, and materials based on Apollo spacecraft heat shields are used to insulate structural steel supports in high-rise buildings.

Lightweight, heat-resistant alloys reduce weight of equipment.

Heat-resistant textiles based on astronaut flight suits

